WOLSTENHOLME NUMBERS AND A GCD CONJECTURE OF ELDAR AND ORDOWSKI: INFINITE FAMILIES, FORMALLY VERIFIED

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Abstract. Let $a(n)$ be the numerator of $H_n^{(2)} = \sum_{k=1}^{n} k^{-2}$ (OEIS A007406). Eldar and Ordowski conjectured (2019) that $\gcd(n, a(n-1)) = 1$ for every composite $n > 3$; for primes $n \geq 5$ Wolstenholme's theorem gives $\gcd = n$. We prove a leading-digit criterion: if q is prime, $q \leq m$, and $q \nmid a(U)$, where U is the leading base-q digit of m, then $q \nmid a(m)$. As consequences, $\gcd(6, a(m)) = 1$ for all $m \geq 1$, and the conjecture holds for all $n = 2^i 3^j > 3$, for all $n = 2q$ ($q \geq 3$ prime), $n = 3q$ ($q \geq 7$), $n = 4q$ ($q \geq 5$, $q \neq 7$), and more generally for $n = cq$ with $c \geq 2$ 3-smooth and $q > c$ a prime not dividing $a(c-1)$, which covers all sufficiently large q for each c; this extends to $n = cq^k$ for every $k \geq 1$ (e.g. $n = 2q^k$). We also formalize Wolstenholme's theorem $p \mid a(p-1)$ for primes $p \geq 5$, so the conjectured *equality* $\gcd(n, a(n-1)) = \mathrm{A089026}(n)$ is verified for every prime $n > 3$ as well. All results are formalized in Lean 4 with Mathlib, stated against the definitions in the `google-deepmind/formal-conjectures` repository, with no `sorry` and only the standard axioms.

1. The conjecture

OEIS A007406 records: *"Conjecture: for $n > 3$, $\gcd(n, a(n-1)) = \mathrm{A089026}(n)$. Checked up to $n = 10^5$."* (A. Eldar, T. Ordowski, Jul. 28, 2019). Here $\mathrm{A089026}(n) = n$ if n is prime and 1 otherwise. J. Sondow added that the prime case follows from Wolstenholme's theorem and that *"it remains to show that* $\gcd(n, a(n-1)) = 1$ *if* $n > 3$ *is composite."* The statement is formalized as `OeisA7406.conjecture` in `FormalConjectures/OEIS/7406.lean`, tagged `research open`.

2. The leading-digit lemma

Throughout, q is prime and v_q is the q-adic valuation on $\mathbb{Q}$.

Lemma 1. *Let $m \geq 1$, $s = \lfloor \log_q m \rfloor$ and $U = \lfloor m/q^s \rfloor$, so $1 \leq U < q$. If $q \nmid a(U)$ then $v_q(H_m^{(2)}) = -2s$.*

Proof. Since $U < q$, every term u^{-2}, $1 \leq u \leq U$, is a q-adic unit, so $v_q(H_U^{(2)}) \geq 0$; as $q \nmid a(U)$ and the denominator of $H_U^{(2)}$ is coprime to q,

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$v_q(H_U^{(2)}) = 0$. Split $\{1, \dots, m\}$ into $S_1 = \{k : q^s \mid k\} = \{q^s u : 1 \le u \le U\}$ and S_0. Then $\sum_{S_1} k^{-2} = q^{-2s} H_U^{(2)}$ has valuation $-2s$. For $k \in S_0$ we have $v_q(k) \le s-1$, so $v_q(k^{-2}) \ge -2s+2$; since all terms are positive, the ultrametric inequality gives $v_q(\sum_{S_0} k^{-2}) \ge -2s+1$. Hence $v_q(H_m^{(2)}) = -2s$. □

Theorem 2 (criterion)**.** *If $q \le m$ and $q \nmid a(\lfloor m/q^{\lfloor \log_q m \rfloor} \rfloor)$, then $q \nmid a(m)$. Consequently, if $n > 3$ is composite and for every prime $q \mid n$, q does not divide a of the leading base-q digit of $n-1$, then* $\gcd(n, a(n-1)) = 1$.

Proof. Here $s \ge 1$, so $v_q(H_m^{(2)}) < 0$; if $q \mid a(m)$ then $v_q(H_m^{(2)}) \ge 1$. For the second claim, a prime divisor q of composite n satisfies $q < n$, so $q \le n-1$. □

This mirrors the classical leading-digit argument for $p \nmid$ numerators of ordinary harmonic numbers (cf. Eswarathasan–Levine, Boyd, and the discussion of Erdős Problem #291).

3. Consequences

Theorem 3. *For all $m \ge 1$,* $\gcd(6, a(m)) = 1$.

Proof. For $q = 2$ the leading digit is 1 and $a(1) = 1$; for $q = 3$ it is 1 or 2, and $a(1) = 1$, $a(2) = 5$. Small $m < q$ are checked directly. □

Corollary 4. *The conjecture holds for every $n = 2^i 3^j > 3$.*

Theorem 5. *Let $c = 2^i 3^j \ge 2$ and let $q > c$ be prime with $q \nmid a(c-1)$. Then* $\gcd(cq, a(cq-1)) = 1$*, that is, the conjecture holds at $n = cq$.*

Proof. Primes $2, 3$ are handled by Theorem 3. For q: $q \le cq - 1 < q^2$, so $s = 1$ and the leading digit is $\lfloor (cq-1)/q \rfloor = c-1$. □

Corollary 6. *The conjecture holds for $n = 2q$ (all primes $q \ge 3$), $n = 3q$ (primes $q \ge 7$), and $n = 4q$ (primes $q \ge 5$, $q \ne 7$), using $a(1) = 1$, $a(2) = 5$, $a(3) = 49$. For each 3-smooth $c \ge 2$ it holds for $n = cq$ for all primes $q > \max(c, a(c-1))$.*

Theorem 7. *Let $c = 2^i 3^j \ge 2$, $q > c$ prime with $q \nmid a(c-1)$, and $k \ge 1$. Then the conjecture holds at $n = cq^k$. In particular it holds for $n = 2q^k$, for every odd prime q and every $k \ge 1$.*

Proof. For $m = cq^k - 1$ we have $q^k \le (c-1)q^k \le m < cq^k < q^{k+1}$, so $\lfloor \log_q m \rfloor = k$ and the leading digit is $c-1$. Primes $2, 3$ are handled by Theorem 3. □

4. The prime direction

Theorem 8 (Wolstenholme)**.** *For every prime $p \ge 5$, $p \mid a(p-1)$. Hence* $\gcd(p, a(p-1)) = p =$ A089026(p) *for every prime $p > 3$.*

Proof. Let $F = (p-1)!$ and $N = \sum_{k=1}^{p-1}(F/k)^2$, so that $H_{p-1}^{(2)}F^2 = N$. In $\mathbb{F}_p$, Wilson gives $k \cdot (F/k) = -1$, so $(F/k)^2 = (F/k)^2 k^{p-1} = k^{p-3}$. Therefore $N \equiv \sum_{x\in\mathbb{F}_p} x^{p-3} = 0$, since $0 < p-3 < p-1$. As $p \nmid F^2$, p divides the numerator. □

Corollary 9. *The full conjecture* $\gcd(n, a(n-1)) = \mathrm{A089026}(n)$ *holds for every* $n > 3$ *that is prime,* 3*-smooth, or of the form* $2q^k$ *(*q *an odd prime,* $k \geq 1$*), and for* $n = cq^k$ *as in Theorem 7.*

Remark 10 (Coverage and limits). A short computation (`coverage.py`) shows the criterion of Theorem 2 settles 76,100 of the 90,407 composites $4 \leq n \leq 10^5$ (about 84%). It cannot settle every case: $q \mid a(q-1)$ (Wolstenholme) and $q \mid a((q-1)/2)$ for all $q \geq 5$, so for instance $n = q^2$ needs a second-order analysis. There $H_{q^2-1}^{(2)} = \sum_{q\nmid k<q^2} k^{-2} + q^{-2}H_{q-1}^{(2)}$, and the first sum is $\equiv 0 \pmod{q^2}$. One finds that $q \mid a(q^2-1)$ would force $v_q(H_{q-1}^{(2)}) \geq 3$, which by Glaisher's congruence $H_{q-1}^{(2)} \equiv \frac{2}{3}qB_{q-3} \pmod{q^2}$ requires $q \mid B_{q-3}$ (a Wolstenholme prime, e.g. 16843) together with a further congruence. We do not formalize this.

5. Formalization

The Lean project `Wolst` (Lean `v4.35.0-rc4`, Mathlib `v4.35.0-rc4`) defines `a` and `A89026` verbatim from `formal-conjectures` and proves: `padicValRat_H` (Lemma 1), `not_dvd_a` and `conjecture_of_leading_digits` (Theorem 2), `coprime_six_a`, `conjecture_smooth`, `conjecture_smooth_mul_prime`, `conjecture_two/three/four_m` `conjecture_smooth_mul_large_prime`, `conjecture_smooth_mul_prime_pow`, `conjecture_two_mul_prime_pow`, `prime_dvd_a` (Theorem 8), `conjecture_prime`, and `conjecture_on_families`. Each conclusion is literally `Nat.gcd n (a (n - 1)) = A89026 n`. `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. No `sorry` or `native_decide` is used.

6. Novelty

As of October 12, 2026, OEIS A007406 lists the conjecture as open, and the formal-conjectures entry is tagged `research open`. A web search found no published proof of any infinite family. The leading-digit technique itself is classical. What we believe is new is applying it to this conjecture and the resulting families.

References

[1] OEIS Foundation, *A007406: Wolstenholme numbers*, `https://oeis.org/A007406`.
[2] Google DeepMind, *formal-conjectures*, `FormalConjectures/OEIS/7406.lean`.
[3] R. Meštrović, *Wolstenholme's theorem: its generalizations and extensions*, arXiv:1111.3057.
[4] A. Eswarathasan, E. Levine, *p-integral harmonic sums*, Discrete Math. 91 (1991).